Yafan Huang

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EDUCATION

Huazhong University of Science and Technology (HUST) Wuhan, China
M.E. in Computer Science *09/2018-06/2021*
School of Computer Science & Technology GPA: 3.64/4.0

Hunan University (HNU) Changsha, China
B.E. in Software Engineering *09/2014-06/2018*
College of Computer Science and Electronic Engineering GPA: 3.36/4.5

PUBLICATIONS

- Feng Zhao, Xiangyu Gui, **Yafan Huang**, Hai Jin, Laurence T. Yang, **Dynamic Entity-based Named Entity Recognition Under Unconstrained Tagging Schemes**, *IEEE Transaction on Big Data, Jun 2020*
- **Yafan Huang**, Feng Zhao, Shihui Song, Xiangyu Gui, Hai Jin, **Path-enhanced Explainable Recommendation with Knowledge Graphs,** *World Wide Web Journal 2021*
- Shihui Song, **Yafan Huang**, Hongwei Lu, **Rumor Detection Algorithm via Spectral Convolution in Directed Graphs**, In submission

RESEARCH EXPERIENCES

Research on Inductive Rumor Detection in Large Graphs via Centrality Measures
Key Member *07/2020-12/2020*
Research Objective: Aim to establish a Graph Neural Network (GNN) based model and provide robust prediction results in rumor detection task

- Designed a novel end-to-end model named OID-GCN (Out-In-Degree Graph Convolution Networks) for accurately representing rumor propagation networks and textual information
- Adopted Katz centrality to differentiate the importance of nodes and further labeled the "Opinion Leaders" in rumor detection task
- Used Twitter-15 and Twitter-16 benchmark datasets to conduct comprehensive experiments, from which the results demonstrated our model achieved a higher performance than existing method like CNN-OM, TF-IDF, GRE-2, LSTM-2

The Construction of Technology Resource Pool via Knowledge Graph (National Key Research & Development Project)
Leader *03/2019- Present*
Research Objective: Aim to build an information management system that includes all the science and technology resource of Hubei, Hunan and Jiangxi province of China

- Implement knowledge graph engine and data analysis functions (including CRUD, Temporal data analysis, Regional resource analysis, and Sub-graph analysis) via Java
- Visualize knowledge graph and data analysis modules via vis.js, ECharts, and Chart.js to realize visualization of large amounts of data from multiple sources
- Design the interface via Bootstrap-based HTML and CSS and deploy the Maven project on Linux

Research on Path-enhanced Explainable Recommendation with Knowledge Graphs
Leader *03/2019- 04/2020*
Research Objective: Aim to enhance the explainability and accuracy of existing knowledge graph-based recommendation algorithms by using Bi-LSTM network and information entropy

- Designed path-enhanced recurrent network (PeRN) model by incorporating entropy-based encoder to a Bi-LSTM network encoder
- Abstracted two real-world recommendation datasets, namely, KKBox and IMDb, into two new Knowledge Graph recommendation datasets
- The result showed that PeRN outperformed several past models in accuracy and has greater ability in reasoning and solving cold-start issues

The Application Research on Dynamic Entity-Based Named Entity Recognition Under Unconstrained Tagging Schemes

Key Member *03/2019- 08/2019*

Research Objective: Aim to use an unconstrained tagging schemes to alleviate features overlapping and improve accuracy of Named Entity Recognition (NER) tasks

- Proposed dynamic entity-based method to extract and classify named entities so as to further exploit word-level and character-level information in NER
- Compared and analyzed two proposed unconstrained tagging schemes with existing schemes, and the results showed the unconstrained tagging scheme performed well in four different language datasets experiment, especially in recognizing entities with long lengths.

INTERNSHIP EXPERIENCES

Product Development Center, Eastman Kodak Company **Shanghai, China**

Software Developer Intern *07/2017-11/2017*

- Adopted Python to develop scripts and test scripts of the Prinergy Version Check which was formally applied by the product development center
- Responsible for design reviews and code reviews for the Prinergy team, had successfully fixed the bugs in the Prinergy Server and Workshop via C++ and Java, and learned more about the programming project specifications
- Learned to use enterprise applications such as managed task by JIRA, managed code by Perforce, and held English meetings via Skype

EXTRACURRICULAR ACTIVITIES

The 7th CCF Conference on Big Data, China Computer Federation **Wuhan, China**

Volunteer *09/2019*

- Welcomed the conference attendees and helped to facilitate the registration process including direction guide, material preparation and technical support on the spot
- Discussed and interacted with researchers on big data topic and answered the questions the attendees may have

The Student Union, Hunan University **Changsha, China**

Vice President *09/2017-06/2018*

- Assisted the president in supervising the overall administration, such as attending monthly officer meetings to ensure communication and collaboration, serving in the construction of internal rules and regulations, and etc.
- Directed outdoor activities for the minority students and encouraged participation, had successfully organized the graduation party in the college with over 1000 students engaged in it
- Provided immediate news report on school website and other social media platforms and maintained media support for school events

The "Internet+" College Students Innovation and Entrepreneurship Contest **Changsha, China**

Key participant *08/2015*

- Actively participated in requirement analysis, UI design with HTML, CSS and Photoshop, and user manual creating in the project "Food Safety Cloud Platform Based on Electronic Intelligent Scale"
- Delivered project presentation and the oral defense, won the second prize in the contest (10 teams out of 3683 teams)

ACHIEVEMENTS/AWARDS

- 1st class Scholarship for Academic Excellence, awarded by HUST, School Level *09/2020*
- 2nd class Scholarship for Academic Excellence, awarded by HUST, School Level *09/2019*
- 1st class Scholarship for Academic Excellence, awarded by HUST, School Level *09/2018*
- 1st class Scholarship for Academic Excellence, award by HNU, School Level *09/2016*
- 2nd Prize in China College Students "Internet+" Innovation & Entrepreneurship Competition, Provincial Level *08/2015*

SKILLS/HOBBIES

- **Programming**: Python, C++, Java, SQL, JavaScript, HTML, CSS, MATLAB, LaTex, Markdown
- **Tools**: Linux, VS Code, IntelliJ IDEA, Git, Jupyter
- **Data Science**: PyTorch, Pandas, Numpy, Matplotlib, Dgl, Common Machine Learning Models
- **Language**: English (Fluent), TOEFL 99 (R28; L23; S23; W25)
- **Hobbies**: good at singing, playing instruments, and musical composition; Enjoying long-distance race and E-sports